Bridging Misconception Detection & Dialogue-Based Tutoring for Programming Education

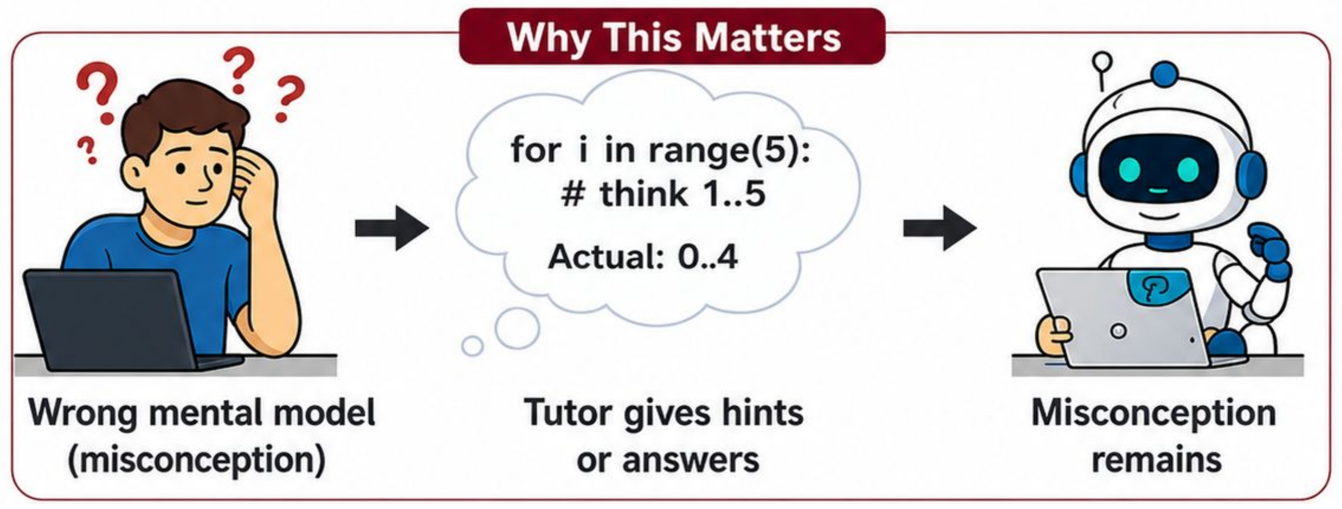
Catherine Awad, Ruiwen Guan, Hanieh Hedayati
Advisor: Dr. Oana Ignat

Santa Clara Engineering 1851
Santa Clara University NLP CLASS SCU

Can misconception detection be integrated directly into tutoring to improve learning and avoid over-tutoring?

Motivation & Methodology

Why This Matters



Wrong mental model (misconception) → Tutor gives hints or answers → Misconception remains

Existing Systems

TRAVER (Dialogue Tutoring)	McMiner (Misconception Detection)
✓ Dialogue-tutoring	✓ Detects misconceptions
✓ Socratic questioning	✓ Finds false beliefs
✓ Adaptive conversation	✓ Execution-based diagnostics
✗ Cannot explicitly detect misconceptions	✗ Not a tutoring system

Neither system alone can both detect misconceptions and correct them through dialogue.

How McMiner Works

Student Code → Pattern Analysis → Infer Misconception → Tutor Feedback

McMiner identifies the specific false belief and informs the tutor.

How TRAVER Works (Dialogue Tutoring)

Student Question or Response → TRAVER Tutor Responds → Continued Dialogue (Socratic Questioning) → Student Improves Over Time

TRAVER guides the student through dialogue to solve the problem.

Experimental Setup

Original Paper (TRAVER)	
Repositories	25
Tasks	100
Tutor Model	Llama 3.1-70B
Student Model	Mixtral-8x7B
Quantization	FP16

→

Our Reproduction	
Repositories	4
Tasks	22
Tutor Model	Llama 3.1-70B
Student Model	Mistral-7B
Quantization	NF4 (4-bit)

Novel idea: Batch injection

TRAVER (Dialogue) → McMiner (Diagnostics) → Many Misconceptions → Overloaded Tutor

1: McMiner
Misconception Detection

Detects misconceptions from code.

Misconception-Aware Tutor

Surfaces and corrects false beliefs through dialogue.

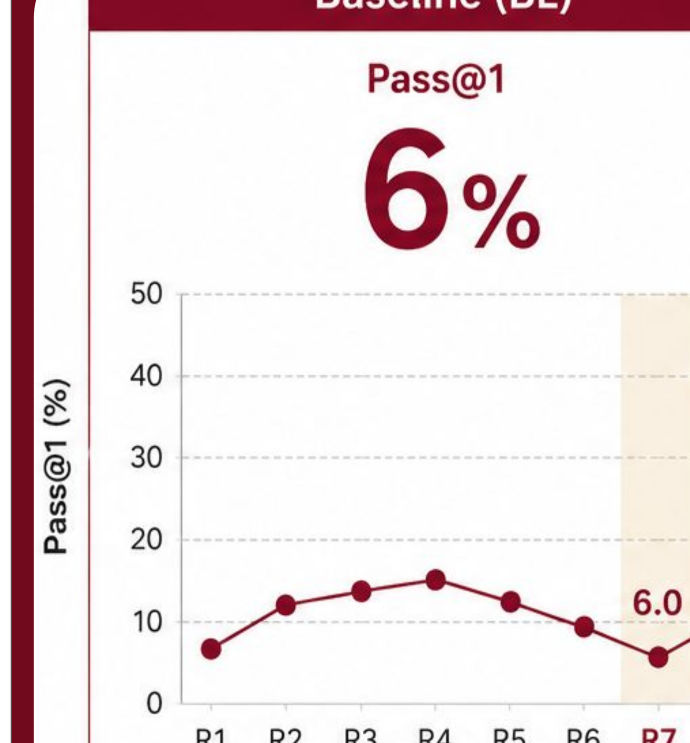
2: TRAVER
Dialogue Tutoring

Engages students in dialogue to guide problem solving.

Headline Result: Easy Volcap - High level (R7)

Baseline (BL)

Pass@1
6%



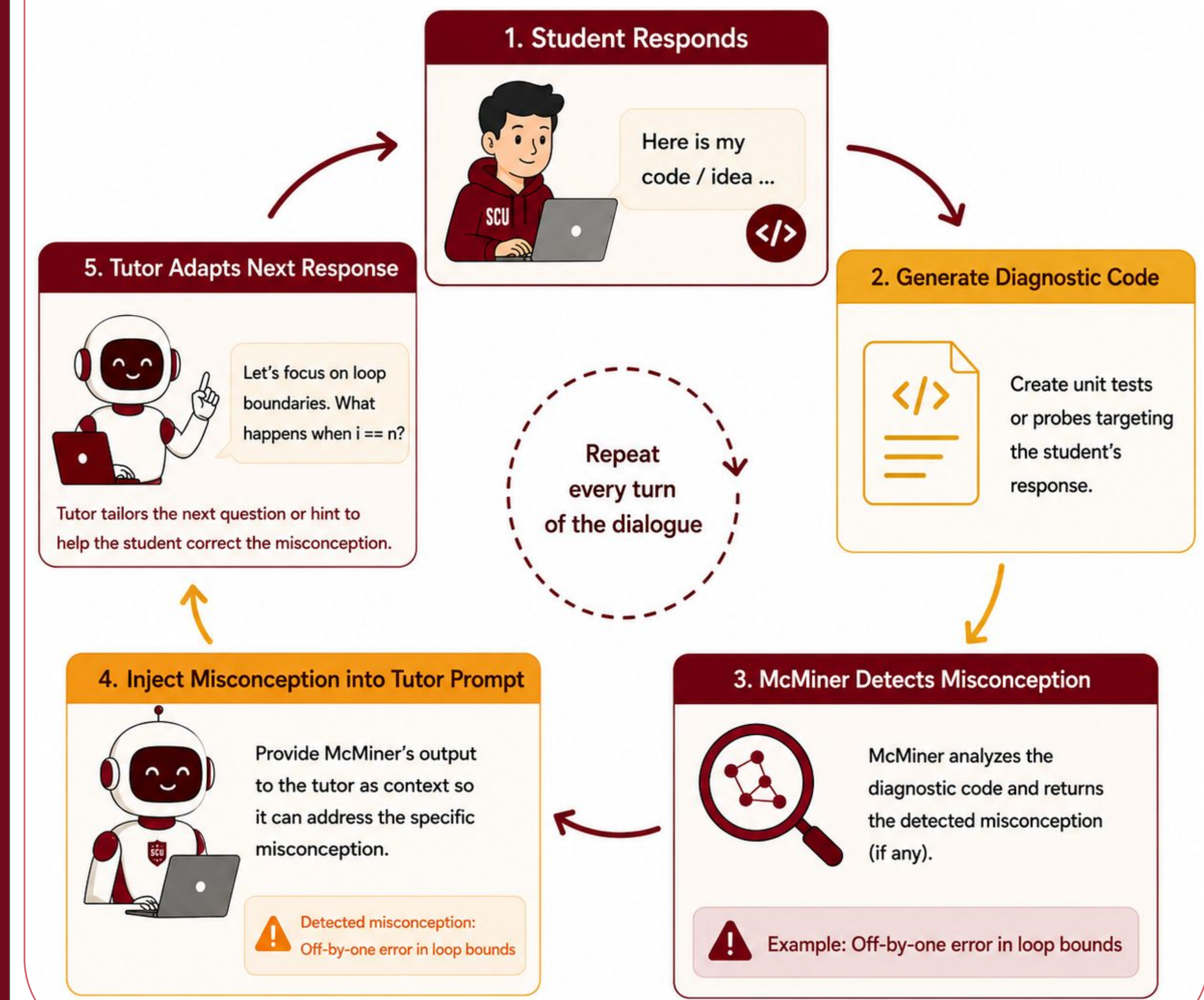
IMPROVEMENT
+25.7
percentage points

McMiner-Clean (CL)

Pass@1
31.7%

Scan Here For Code, Paper and Demo

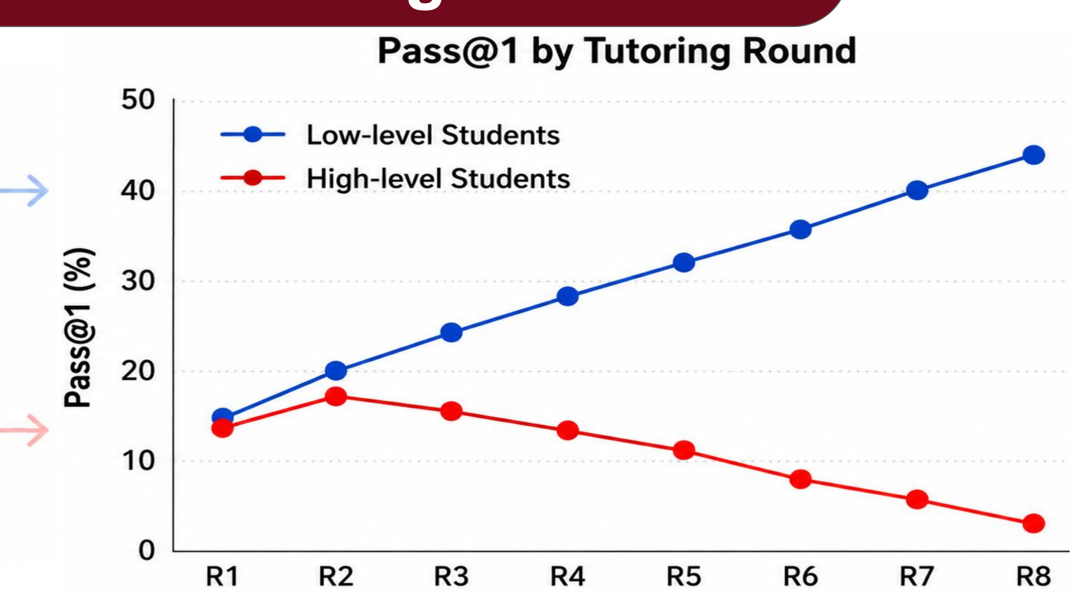
McMiner In The Loop (Our Approach)



Over Tutoring Problem

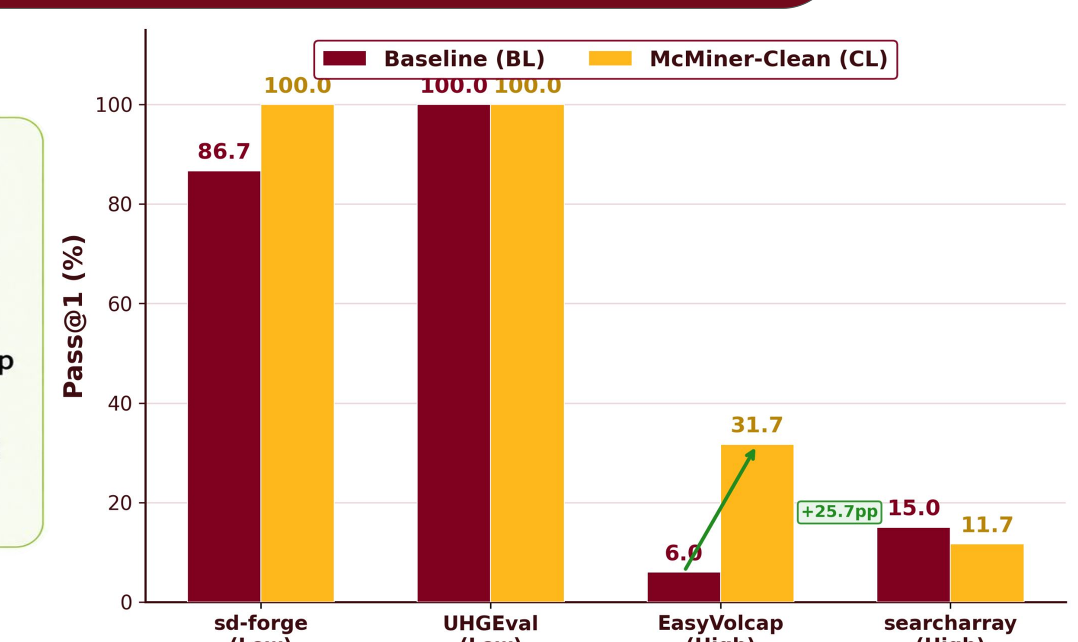
Low-level students: More tutoring helps.

High-level students: More tutoring hurts.



Overall Results

- ✓ Improves UHGEval
- ✓ Improves sd-forge
- ✓ Improves EasyVolcap
- ✓ Solves over-tutoring problem



Limitations

- Reduced benchmark size (limited repositories)
- Smaller student model (limited capacity)
- Limited to coding tasks (Python-focused)
- Results may vary by domain and dataset

These limitations motivate ongoing improvements and future work.

Future Work

- Larger-scale evaluations across more repositories and domains
- Real student studies to measure learning gains
- Adaptive tutoring policies based on student states and histories
- Extend beyond Python to other languages and tasks

We aim to build more robust, generalizable tutoring systems.